Report: Data Analysis and Vendor Performance Evaluation

1. Executive Summary

The objective of this project was to evaluate vendor performance based on various key metrics, using data from multiple sources. Through data cleaning, exploratory analysis, and visualization techniques, key performance insights were derived to help identify strengths and improvement areas in the vendors' operations. The project culminated in the creation of an interactive dashboard, making it easier to track vendor performance over time.

2. Objective

The goal of the analysis was to assess the performance of vendors based on transaction data, sales-to-purchase ratio, and other related metrics. The insights were used to recommend improvements for underperforming vendors and highlight top performers.

3. Data Collection

The data for this analysis was collected from a CSV file containing information about vendor performance. Key data points include:

- **VendorNumber**: Unique identifier for each vendor.
- **Sales to Purchase Ratio**: Performance metric.
- **Other transactional data**: Sales figures, purchase amounts, delivery times, etc.

4. Data Preparation

Data cleaning and preprocessing included:

- **Handling Missing Values**: Imputation used to fill missing data.
- **Normalization**: Applied to certain fields to ensure comparability.
- **Merging Datasets**: Combined multiple sources to create a unified dataset for analysis.

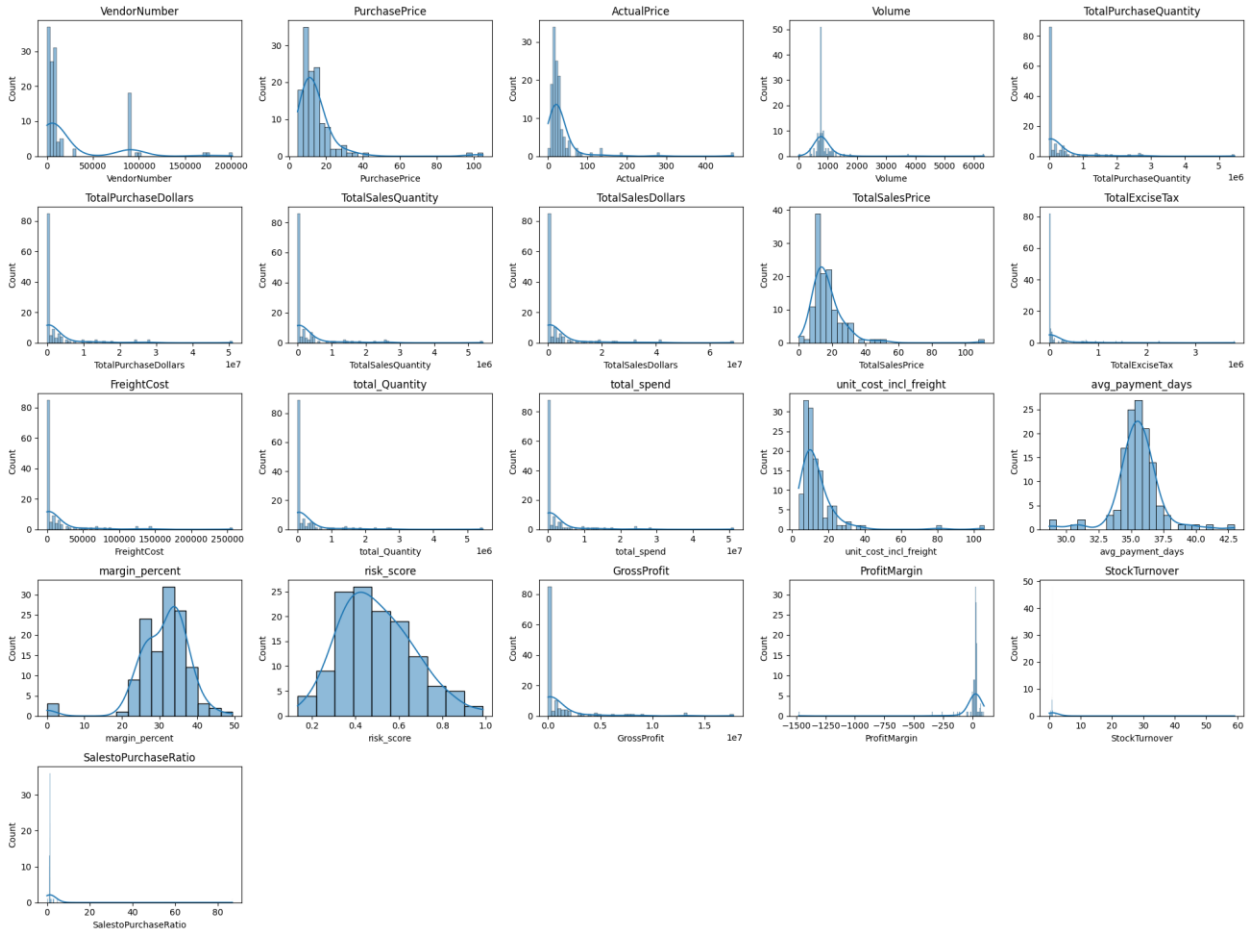
Data Summary Statistic:

	count	mean	std	min	25%	50%	75%	max
VendorNumber	129.0	2.337027e+04	3.975154e+04	2.0000	2555.000000	7153.000000	1.156700e+04	2.013590e+05
PurchasePrice	129.0	1.469027e+01	1.279596e+01	4.5400	8.510100	11.776900	1.609710e+01	1.050700e+02
ActualPrice	129.0	3.261944e+01	5.137160e+01	0.0000	14.016300	19.990000	2.999000e+01	4.706775e+02
Volume	129.0	8.625871e+02	5.901498e+02	0.0000	734.600000	753.330000	8.398400e+02	6.350000e+03
TotalPurchaseQuantity	129.0	2.853020e+05	7.119085e+05	1.0000	1847.000000	12798.000000	1.926420e+05	5.459788e+06
TotalPurchaseDollars	129.0	2.749691e+06	6.871616e+06	17.0000	25961.040000	146839.550000	1.882833e+06	5.095980e+07
TotalSalesQuantity	129.0	2.795854e+05	7.008918e+05	0.0000	1488.000000	12430.000000	1.907790e+05	5.422703e+06
TotalSalesDollars	129.0	3.881625e+06	9.568160e+06	0.0000	35508.520000	190134.630000	2.646326e+06	6.874242e+07
TotalSalesPrice	129.0	1.749359e+01	1.172774e+01	0.0000	11.511300	15.017000	1.980980e+01	1.112687e+02
TotalExciseTax	129.0	1.534896e+05	4.582217e+05	0.0000	417.020000	5327.700000	5.115563e+04	3.798655e+06
FreightCost	129.0	1.403481e+04	3.506476e+04	0.0900	130.090000	778.250000	9.621580e+03	2.570321e+05
total_Quantity	129.0	2.603440e+05	6.814613e+05	1.0000	1629.000000	9277.000000	1.864640e+05	5.459788e+06
total_spend	129.0	2.508072e+06	6.541438e+06	17.0900	24679.310000	129544.400000	1.724458e+06	5.121683e+07
unit_cost_incl_freight	129.0	1.354846e+01	1.194036e+01	3.5361	7.950800	10.503200	1.471750e+01	1.055500e+02
avg_payment_days	129.0	3.551016e+01	1.751971e+00	28.6700	34.840000	35.550000	3.625000e+01	4.300000e+01
margin_percent	129.0	3.114845e+01	7.073975e+00	0.0000	27.220000	32.800000	3.517000e+01	4.939000e+01
risk_score	129.0	4.995426e-01	1.683492e-01	0.1330	0.372000	0.482000	6.040000e-01	9.870000e-01
GrossProfit	129.0	1.131934e+06	2.732626e+06	-9194.0300	5220.130000	38066.880000	8.195581e+05	1.778262e+07
ProfitMargin	129.0	-inf	NaN	-inf	16.558682	27.156665	3.362976e+01	9.884700e+01
StockTurnover	129.0	1.432127e+00	5.128005e+00	0.0000	0.873202	0.969036	9.897419e-01	5.900000e+01
SalestoPurchaseRatio	129.0	2.064633e+00	7.540312e+00	0.0000	1.198447	1.372809	1.506699e+00	8.673000e+01

Insight:

This summary statistics table provides an overview of the key metrics for the dataset, showcasing the distribution of values for each variable.

- VendorNumber: The count of vendors is 129, with a mean of 23,370. The minimum value is 2, and the maximum value is 2,013,590, indicating a wide range in vendor numbers.
- Purchase Price: The mean is around $14.69, with a minimum of $4.45 and a maximum of $105.70, suggesting some variation in the purchase prices across vendors.
- Total Purchase Dollars: The total purchase dollars range significantly, with a mean of $25.69M, showing a wide distribution from a minimum of $17.07K to a maximum of $55.5M.
- Gross Profit: The mean gross profit is $11.31M, but there's a high variation with a standard deviation of $23.73M. Some vendors show large losses or profits, suggesting diverse business models.
- Risk Score: The mean risk score is 0.43, with a minimum of 0.13 and a maximum of 1.00, indicating a balanced distribution of risk across vendors.
- Stock Turnover: The mean stock turnover is 1.43, with a standard deviation of 5.12, showing that stock turnover varies significantly among vendors.



Insight:

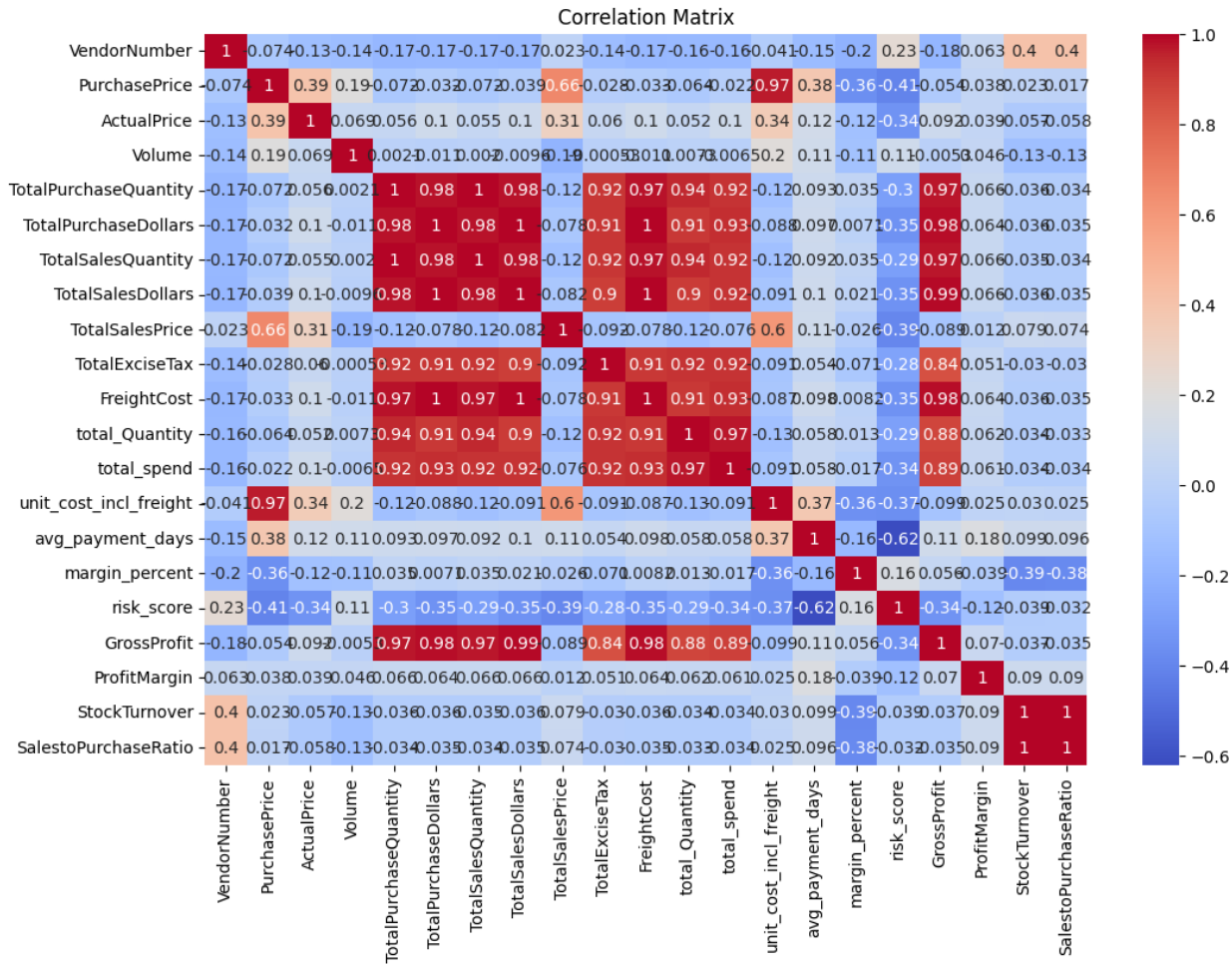
- Right Skewness: The variables PurchasePrice, ActualPrice, Volume, TotalPurchaseQuantity, TotalSalesQuantity, and TotalSalesPrice are right-skewed, meaning that there are a few transactions with very high values, but most transactions are in a small range.
- Negative Profits: There are some negative values in GrossProfit and ProfitMargin, indicating underperforming vendors.
- Risk Score: The Risk_score distribution is relatively uniform, with most vendors falling in the range of 0.2-0.6.
- Sales to Purchase Ratio: The SalesToPurchaseRatio is highly right-skewed, suggesting that most vendors have a low sales-to-purchase ratio.

Recommendation:

- Vendors with negative profit margins and high sales-to-purchase ratios should focus on process optimization and cost-cutting measures.
- For vendors with low sales-to-purchase ratios, improving sales strategies and marketing efforts should be a priority.

- High-volume vendors should be given special attention for enhancing supply chain efficiency and cost management.
- Continuous real-time data monitoring through dashboards, such as Power BI, should be maintained for better decision-making.

Correlation Insights:



Insight:

The correlation matrix reveals strong positive correlations between purchase quantity, purchase dollars, sales quantity, and sales dollars. Specifically, the correlation between TotalPurchaseQuantity and TotalSalesQuantity is very high (0.98), indicating that higher purchase quantities tend to lead to higher sales quantities. Similarly, the relationship between TotalPurchaseDollars and TotalSalesDollars is also strong (0.92), suggesting that more expensive purchases contribute significantly to higher sales revenue.

This confirms the observation that higher purchasing volumes generally lead to increased sales and revenue generation.

5. Methodology

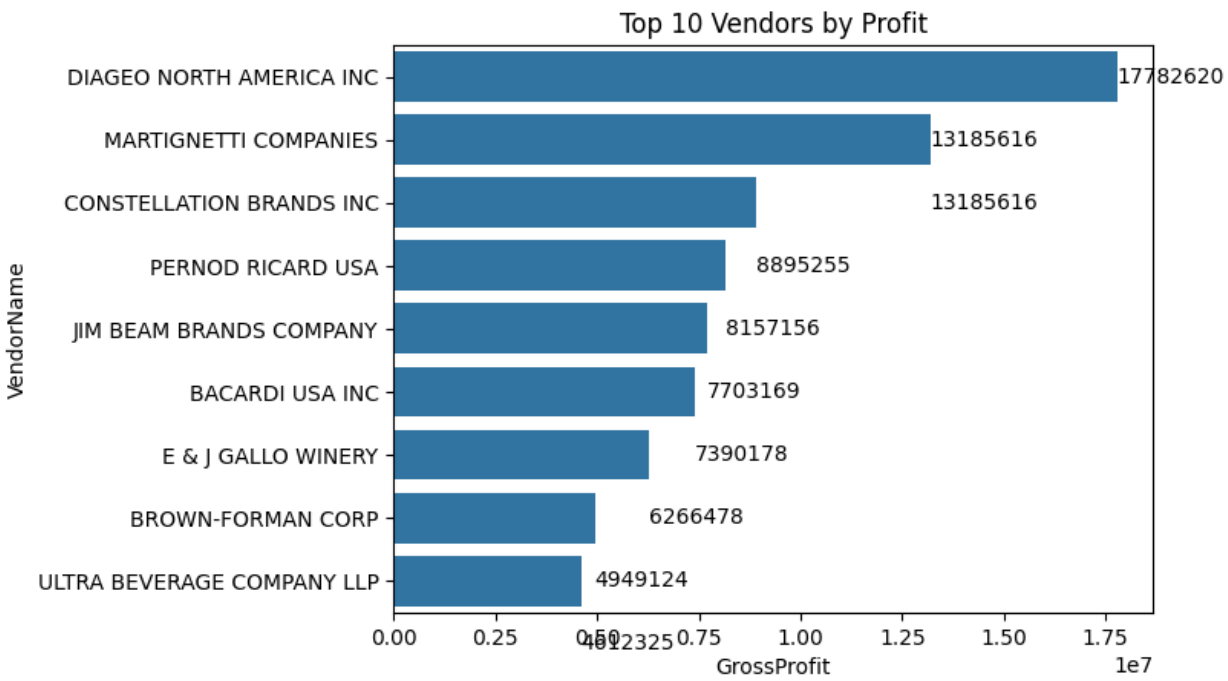
The analysis was carried out using the following methods:

- **Exploratory Data Analysis (EDA)**: Visualizations were used to identify relationships and trends in the data.
- **Python Libraries**: Pandas for data manipulation, Matplotlib and Seaborn for visualization.
- **Power BI**: Dashboard creation to visualize real-time data trends and vendor comparisons.

6. Key Findings

Count of vendors is **129**, with a **mean** of **23,370**. The **minimum** is **2** and **maximum** is **2,013,590**, indicating that there are a wide range of vendor sizes, from smaller vendors to large ones.

- **Top 10 Vendor wise profit:**

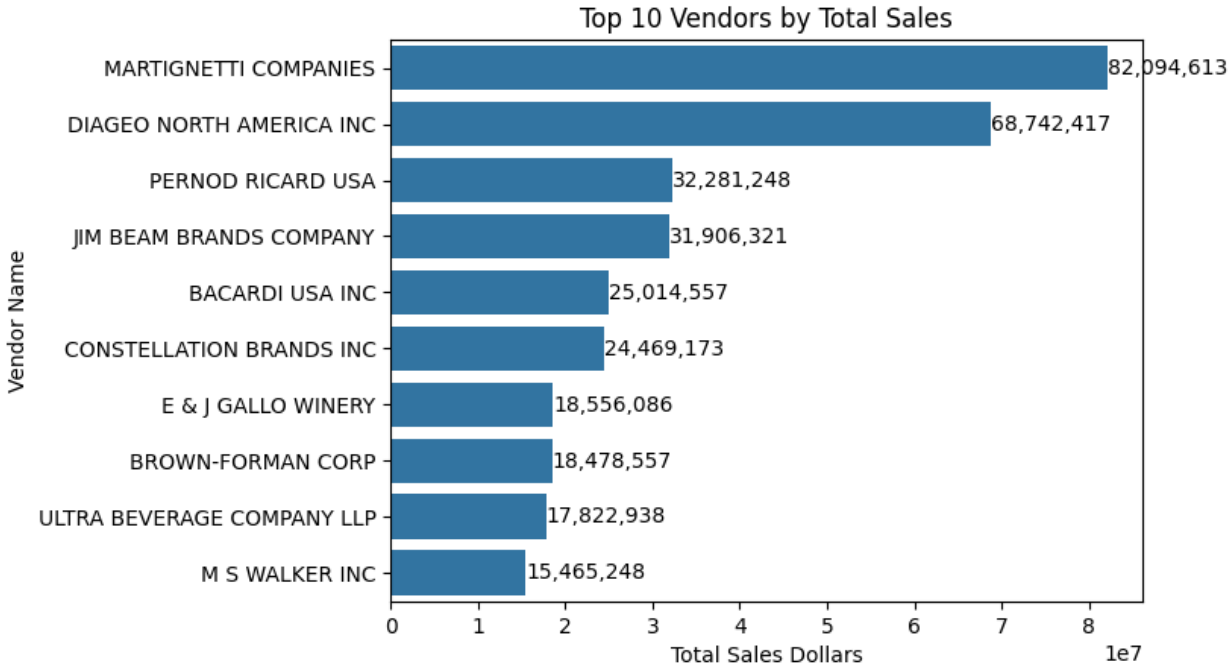


Vendor Contribution:

- DIAGEO NORTH AMERICA INC has the highest profit of $17,782,620, significantly outperforming the other vendors.
- MARTIGNETTI COMPANIES and CONSTELLATION BRANDS INC each have profits of $13,185,616, contributing substantially to overall gross profits as well.
- ULTRA BEVERAGE COMPANY LLP, with the lowest profit in the top 10 at $4,949,124, still plays a notable role in the overall sales.

This chart clearly highlights the vendor-wise contribution to profits, where the top 5 vendors dominate the share, and others, though contributing less, are still part of the overall business.

- **Vendor Comparison**:
  - A bar chart comparing the performance of top vendors in terms of sales.

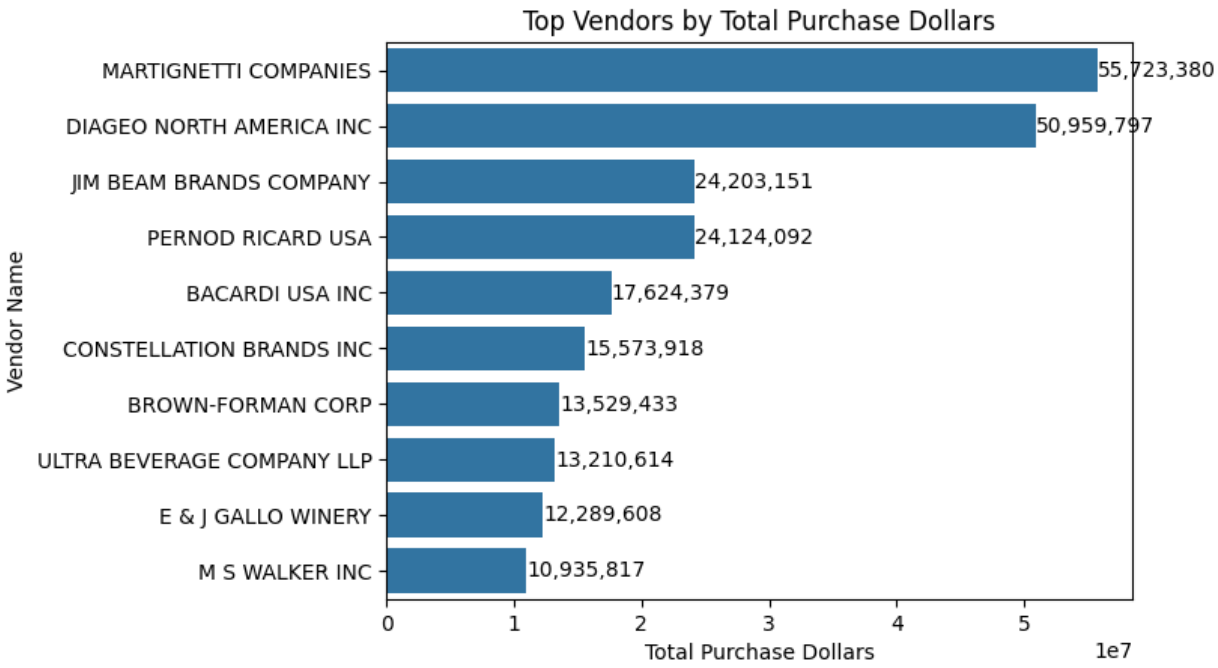


Top 10 Vendors by Total Sales:

- MARTIGNETTI COMPANIES has the highest total sales at $82,994,613, significantly leading the group.
- DIAGEO NORTH AMERICA INC follows closely with $68,742,417 in total sales, making it the second highest contributor.
- PERNOD RICARD USA and JIM BEAM BRANDS COMPANY have total sales of $32,281,248 and $31,906,321, respectively, showing strong performance in the market.
- M S WALKER INC, with the lowest sales in the top 10 at $15,465,248**,** still holds a notable position in terms of total sales.

This chart highlights the top-performing vendors based on total sales and provides a clear view of the vendors' financial contributions.

- **Top Vendors by Total Purchase Dollars**:

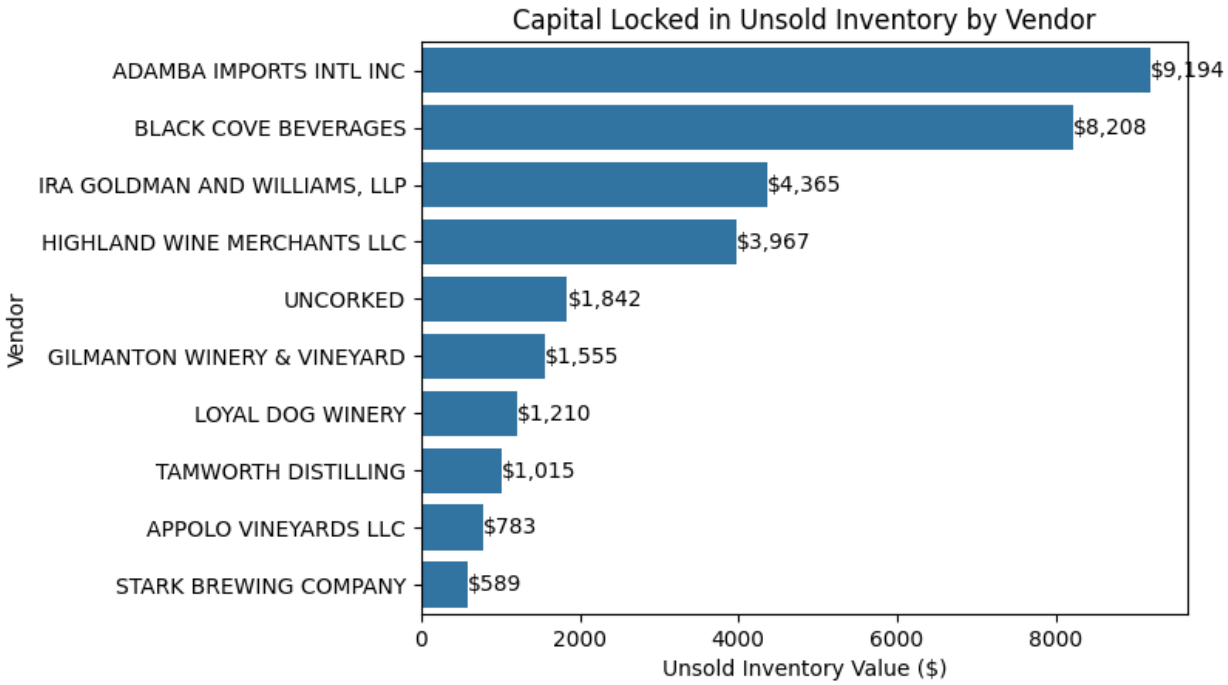


Insight:

Top Vendors by Total Purchase Dollars:

- MARTIGNETTI COMPANIES leads with the highest total purchase dollars of $55,723,380**,** significantly outpacing the others.
- DIAGEO NORTH AMERICA INC follows closely with $50,959,797, contributing substantially to the total purchase amount.
- JIM BEAM BRANDS COMPANY and PERNOD RICARD USA each have total purchases of approximately $24 million, indicating their strong market position**.**
- M S WALKER INC, with the lowest total purchase dollars at $10,935,817**,** still holds a spot in the top 10.

- **Capital Locked in Unsold Inventory by Vendor:**



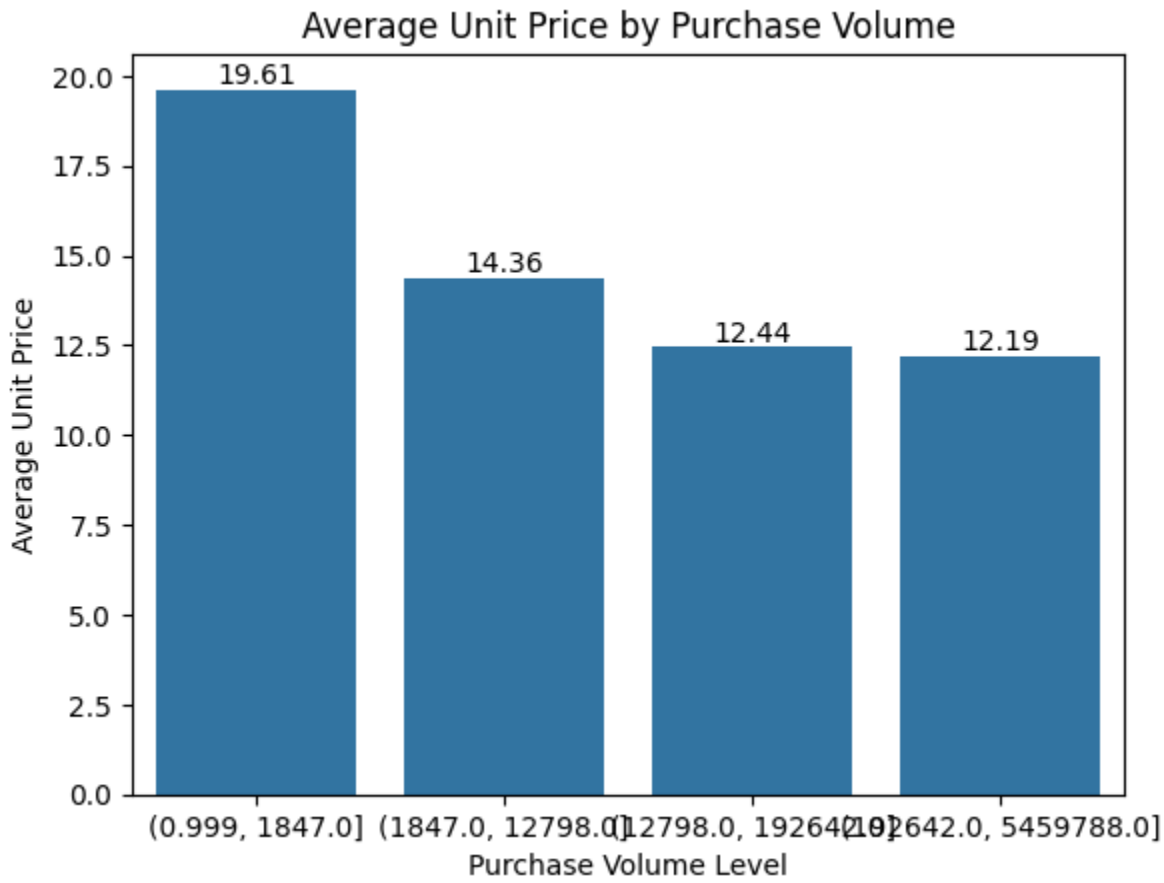
Insight:

Capital Locked in Unsold Inventory by Vendor:

- ADAMBA IMPORTS INTL INC has the highest capital locked in unsold inventory at $9,194, indicating a significant amount of unsold stock.
- BLACK COVE BEVERAGES follows closely with $8,208 in unsold inventory value.
- IRA GOLDMAN AND WILLIAMS, LLP has $4,365, suggesting a moderate amount of unsold inventory.
- STARK BREWING COMPANY has the lowest unsold inventory value at $589, but still contributes to the total unsold capital.

This chart reveals which vendors have the most capital tied up in unsold inventory, helping identify potential issues with inventory turnover.

- **Average Unit Price by Purchase Volume:**



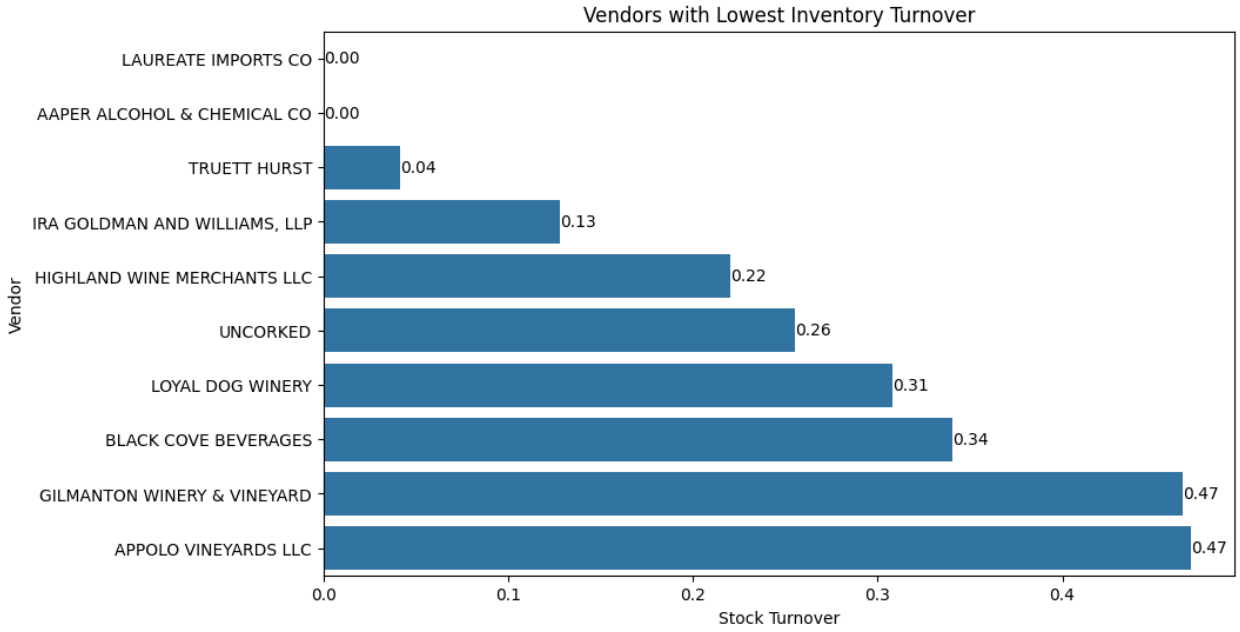
Insight:

The Average Unit Price by Purchase Volume chart shows how the average unit price varies based on different levels of purchase volume.

- **Highest Average Unit Price**: The highest average unit price ($19.61) corresponds to the lowest purchase volume level (0.999 to 1,847 units)**.** This suggests that smaller purchase volumes are associated with higher unit prices.
- **Lower Average Unit Prices**: As the purchase volume increases, the average unit price decreases, with levels around $12.44 to $12.19 for higher purchase volumes, indicating a price reduction as more units are bought.

This chart helps identify the pricing strategy based on purchase volume, showing that vendors offering lower prices are likely to encourage larger purchases. It also reveals that smaller orders come at a premium price.

- **Low inventory turnover:**



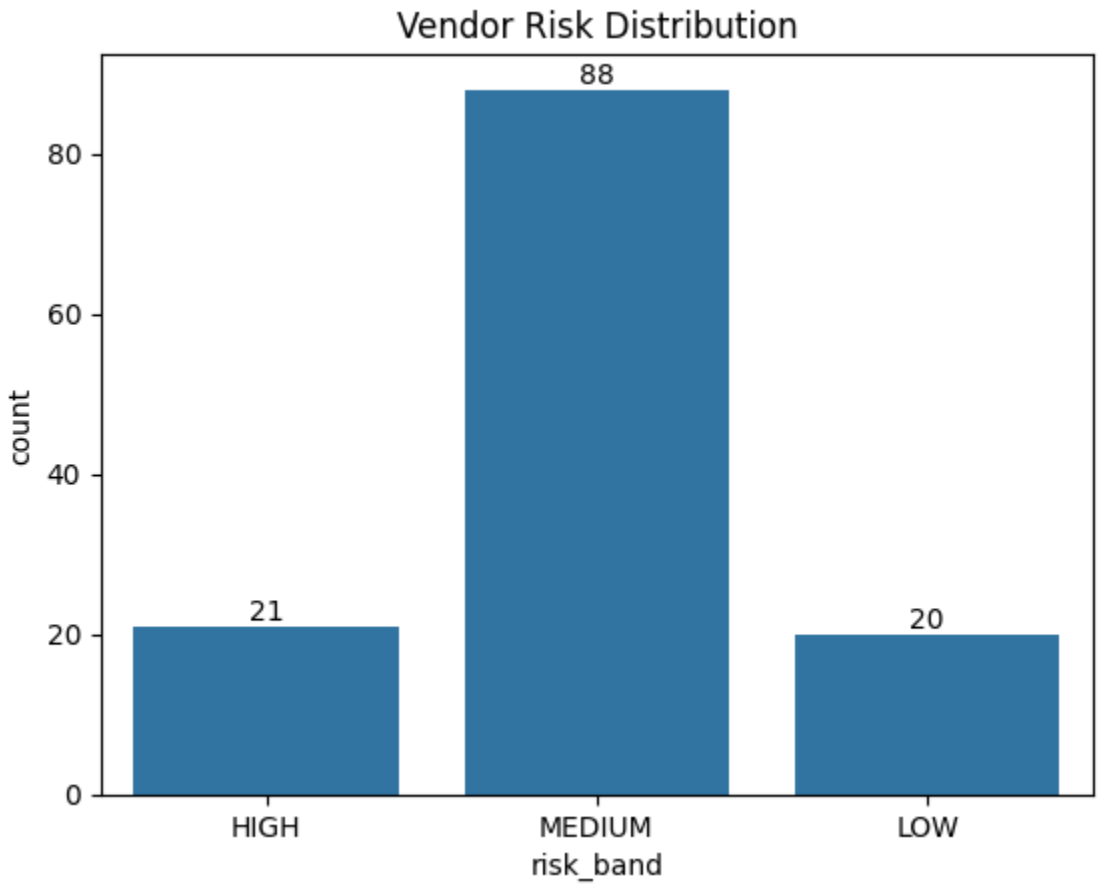
Insight:

Vendors with Lowest Inventory Turnover:

- LAUREATE IMPORTS CO and AAPER ALCOHOL & CHEMICAL CO have the lowest inventory turnover at 0.00, indicating that they are holding onto their stock for an extended period.
- TRUEET HURST follows with a turnover of 0.04, suggesting that it moves its inventory slowly.
- Vendors such as IRA GOLDMAN AND WILLIAMS, LLP and HIGHLAND WINE MERCHANTS LLC have slightly higher turnovers, with values of 0.13 and 0.22, but they still exhibit slower turnover compared to other vendors.

This chart highlights the vendors with low inventory turnover, indicating possible inefficiencies in their inventory management or low sales.

- **Vendor Risk Distribution:**



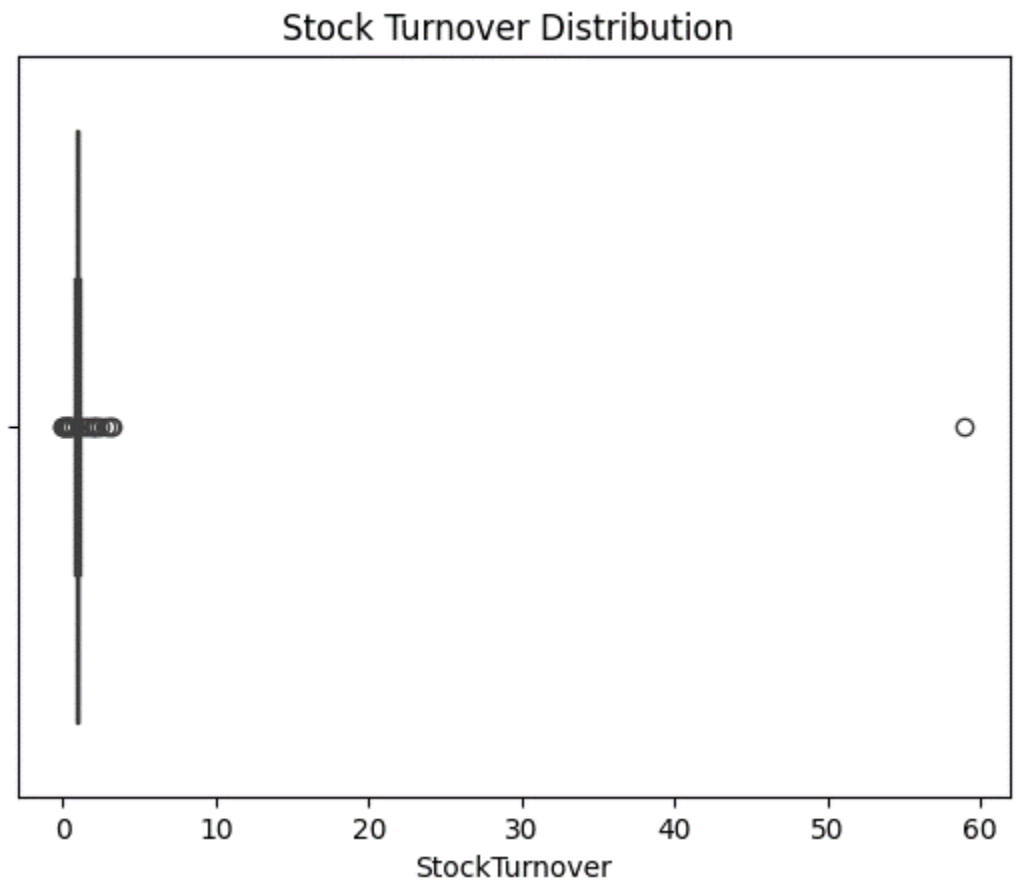
Insight:

Vendor Risk Distribution:

- **MEDIUM Risk**: The dataset contains 88 vendors in the medium-risk category, which forms the largest group.
- **HIGH Risk**: There are 21 vendors in the high-risk band, requiring more attention and improvement.
- **LOW Risk**: 20 vendors fall into the low-risk category, indicating relatively stable performance.

7. Visualizations and Dashboard

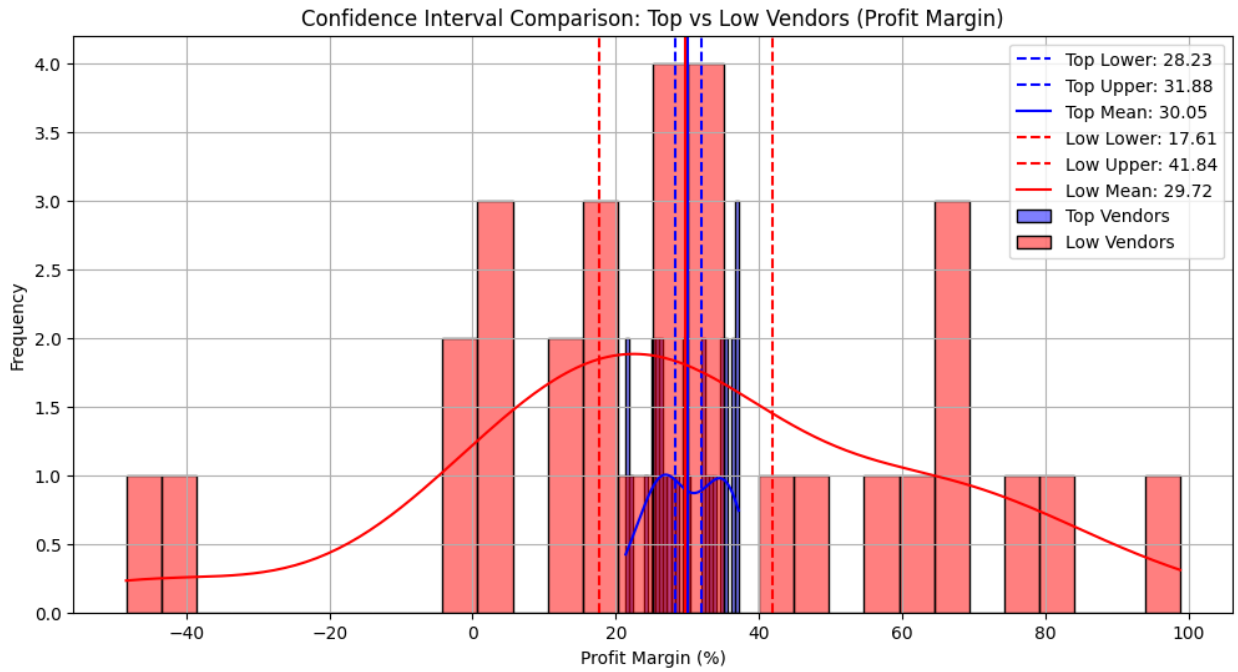
- **Vendor Performance Distribution**:



Vendor Performance Distribution:

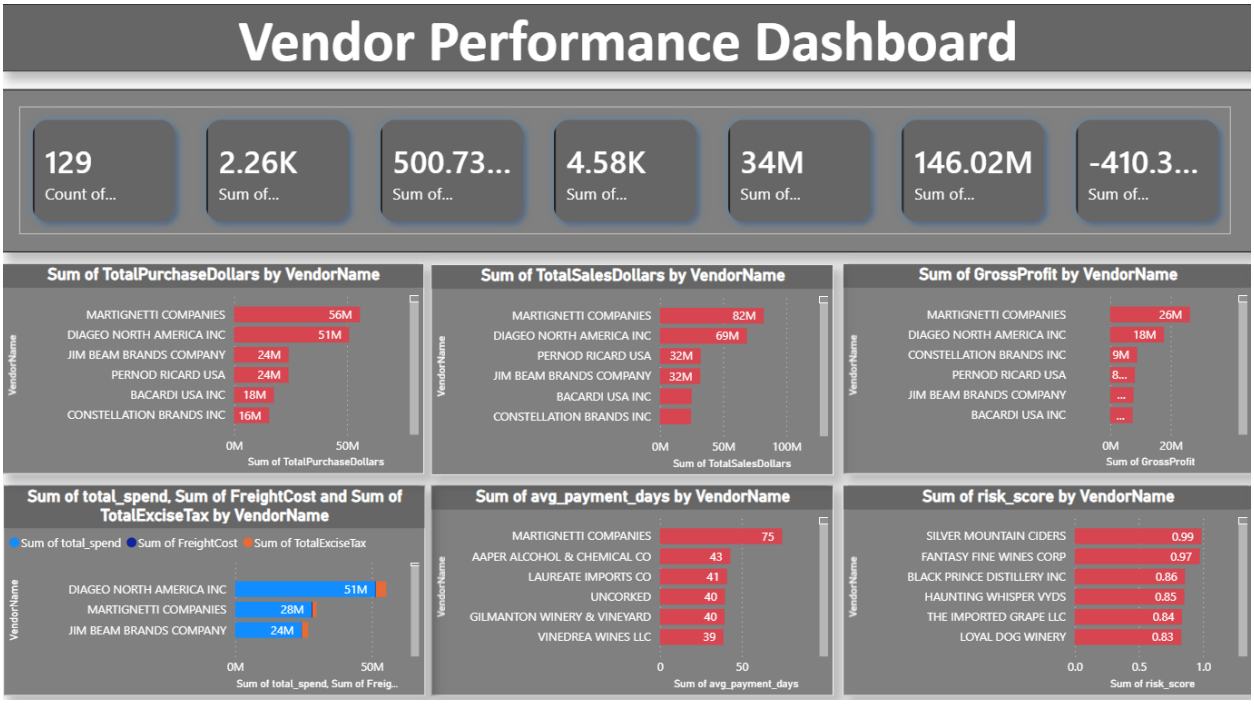
- **Stock Turnover Distribution**: A boxplot has been created to visualize the distribution of stock turnover across vendors. It shows the central tendency of the data, with a few outliers that suggest unusually high stock turnover values for certain vendors.

- **Confidence Interval Comparison: Top vs Low Vendors (Profit Margin)**



Insight:
Although top and low vendors have nearly similar average profit margins, their performance differs significantly in terms of consistency. Top vendors show a narrow confidence interval, indicating stable and reliable performance. In contrast, low vendors have a much wider range, suggesting high variability and greater risk. This means top vendors are more predictable, while low vendors are less consistent and more uncertain.

- **Interactive Dashboard**:
    - A Power BI dashboard was created to provide an interactive platform for real-time analysis.



Insight from Vendor Performance Dashboard:

The Vendor Performance Dashboard provides a clear overview of key metrics such as Total Purchases, Total Sales, Gross Profit, Average Payment Days, and Risk Score for each vendor. Here's a summary of the insights based on the data shown:

- Total Purchase Dollars:
    - MARTIGNETTI COMPANIES leads with $56M in total purchases, followed by DIAGEO NORTH AMERICA INC with $51M.
- Total Sales Dollars:
    - MARTIGNETTI COMPANIES again leads with $82M in total sales, with DIAGEO NORTH AMERICA INC at $69M.
- Gross Profit:
    - MARTIGNETTI COMPANIES has the highest gross profit of $26M, showing strong profitability compared to other vendors.
- Average Payment Days:
    - MARTIGNETTI COMPANIES has the longest average payment days at 75 days, followed by AAPER ALCOHOL & CHEMICAL CO at 43 days, suggesting a slower payment process for some vendors.
- Risk Score:
    - The highest risk score is for SILVER MOUNTAIN CIDERS with 0.99, followed by FANTASY FINE WINES CORP at 0.97, indicating higher risk vendors.

- **Hypothesis Testing:**

Hypothesis

We compare the profit margin between **Top Vendors** and **Low Vendors**.

Null Hypothesis (H_0):
There is no significant difference in profit margin between top vendors and low vendors.
$$H_0 : \mu_{top} = \mu_{low}$$

Alternative Hypothesis (H_1):
There is a significant difference in profit margin between top vendors and low vendors.
$$H_1 : \mu_{top} \neq \mu_{low}$$

8. Conclusion

This analysis has highlighted key performance metrics for vendors, such as the sales-to-purchase ratio and transaction volume. By focusing on the identified weak areas like sales ratio and delivery times, vendors can improve their performance. The interactive dashboard offers a user-friendly interface for tracking these metrics over time.

9. Recommendations

- **Underperforming Vendors**: Improve sales strategies, especially for vendors with low sales-to-purchase ratios.
- **Top Performing Vendors**: Maintain high levels of performance and consider scaling operations for further growth.
- **Further Analysis**: Predictive analytics could be incorporated to forecast vendor performance in future periods.

10. Appendix

- **Data Sources**:
  - Vendor performance data from CSV file (vendor_scorecard_final.csv).
- **Python Code**:
  - Code snippets used for data analysis (if necessary, include code in the appendix).